How Thin Films Grow

A Monte Carlo study of film roughness and trench voids, from random deposition to a process window for void-free filling

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Abstract

This study uses progressively richer Monte Carlo models to examine how surface mobility and feature geometry control thin-film roughness and trench filling. Random deposition gives a growth exponent $\beta = 0.503$, compared with the exact value 1/2. Adding surface relaxation gives $\beta = 0.248$ and a saturation exponent $\alpha = 0.53$, approximating the expected 1/4 and 1/2 scalings. Temperature and deposition-rate sweeps show that increasing the number of surface hops before burial generally reduces roughness. In the trench model, high aspect ratio can prevent void-free filling across the tested process window. A reference recipe for a 16×16-cell trench at 420 K and 1 ML/s passes 24 of 30 simulated wafers; the flat-film reference passes 30 of 30. These results illustrate why process windows should be assessed with repeated runs and explicit acceptance criteria. The lattice, illustrative energy barriers and capped diffusion limit the findings to this model rather than a material-specific manufacturing process.

1 Introduction

Every chip is built from films a few nanometres to a few hundred nanometres thick, stacked dozens of layers deep. They are deposited by sending atoms at a wafer (sputtering, evaporation) and letting them settle. Two things decide whether a film is usable:

1. **Roughness.** A rough film scatters electrons, makes the next layer uneven, and varies in thickness across the wafer. It is measured as **Rq**, the root-mean-square deviation of the surface height from its mean. In this report Rq is the same quantity physicists call the interface width **W**.
2. **Filling of features.** Wires and contacts are trenches and holes etched into the chip and then filled with metal. If the film closes over the top of a trench before the bottom has filled, it seals in an empty pocket, a **void**. Voids raise resistance and cause failures. The standard measure of how well a film reaches the bottom is **step coverage**: film thickness at the bottom of the trench divided by thickness on top.

Both depend on a race between two rates: how fast new atoms arrive, and how fast atoms already on the surface can move to better positions. This report builds that race up one ingredient at a time, checking each step against known theory before adding the next.

The study addresses three questions: whether the simplest models recover their known growth laws; how temperature and deposition rate alter roughness; and how trench geometry changes the probability of meeting a filling specification. The relaxation results are interpreted in the Edwards–Wilkinson framework (Edwards and Wilkinson 1982) and the finite-size scaling framework of Family and Vicsek (Family and Vicsek 1985).

2 Models and experimental design

All models are **Monte Carlo** simulations: atoms arrive one at a time at random positions, and their behaviour is decided by random draws with physically motivated probabilities. Every run uses a fixed random seed, so all results here can be regenerated exactly (`scripts/run_phase{1..4}.py`, `scripts/run_por_yield.py`).

Phase	Model	What an atom does	Question
1	Random deposition	Lands in a random column and stays	Is the code right? (Exact solution exists)
2	Surface relaxation	Rolls into a lower neighbouring column	What does the simplest smoothing do?
3	Thermal hopping	Hops sideways with probability e^(−n · E_bond/kT)	How do temperature and rate set roughness?
4	Trench fill (2D)	Sticks on first contact, then hops toward well-bonded sites	When does a trench trap a void?

The validation experiments average 30 random-deposition runs on 200 columns and 20 relaxation runs at each substrate width (25, 50, 100 and 200 columns). The trench process window averages eight runs per grid cell. Recipe acceptance is assessed separately over 30 seeded, single-wafer simulations, using the same criteria as the interactive application. Sweeps retain their recorded settings and seeds in `results/`; no experimental measurements are used.

The 1D models describe a cross-section of a flat film as a row of columns; `heights[i]` is how many atoms are stacked in column i. Boundaries are periodic, so the row behaves like a ring with no edges.

3 Phase 1: random deposition, a validation baseline

Each atom picks a column uniformly at random and sticks on top of it. Columns never interact, so after t atoms each column's height is a binomial random number and the width follows exactly

$$W(t) = \sqrt{\theta\,(1 - 1/N)}, \qquad \theta = t/N \text{ monolayers.}$$

On a log–log plot this is a straight line of slope $\beta = 1/2$. The simulation (200 columns, 30 runs averaged) gives $\beta = 0.5029 \pm 0.0005$ and matches the exact curve to within 5% at every time. This confirms that the deposition loop, the roughness measurement and the fitting code are correct.

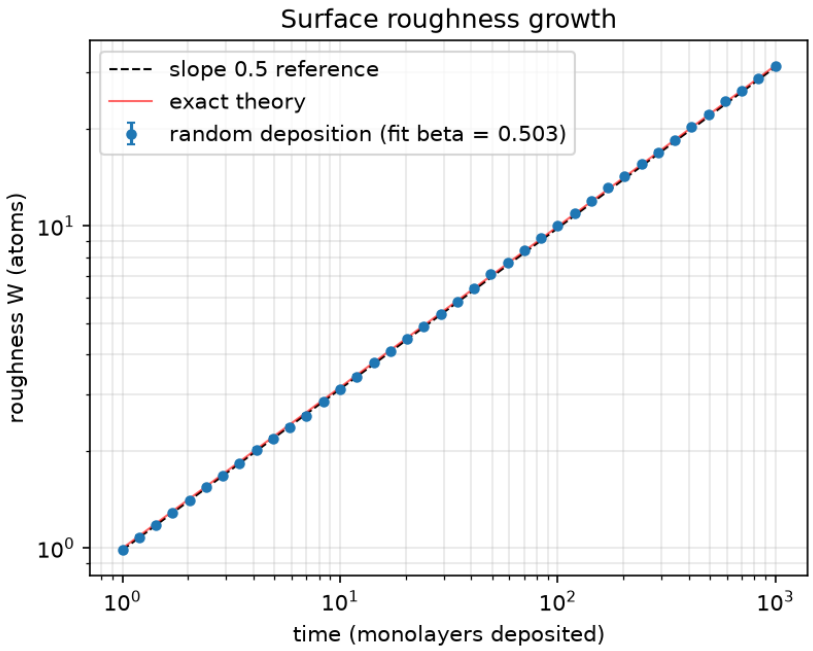


Figure 1: Random deposition: roughness grows as t^1/2 and never stops.

The width **never saturates**. Because no column is affected by its neighbours, nothing limits how far apart neighbouring heights can drift. Real films do not behave like this, which is the motivation for Phase 2.

4 Phase 2: surface relaxation

Now a landed atom checks its two neighbours and rolls into a lower one (picking at random if both are lower). This one local rule is the simplest form of surface diffusion, and it changes the large-scale behaviour completely.

- **Slower growth:** $\beta = 0.248$ (theory 1/4), fitted over the first 100 monolayers on 200 columns.
- **Saturation:** once height differences have spread across the whole substrate, the width stops growing. The plateau scales as $W_{\text{sat}} \propto L^{\alpha}$ with $\alpha = 0.53$ (theory 1/2), and the time to reach it as L^2. Rescaling time by L^2 and width by $L^{1/2}$ collapses all four

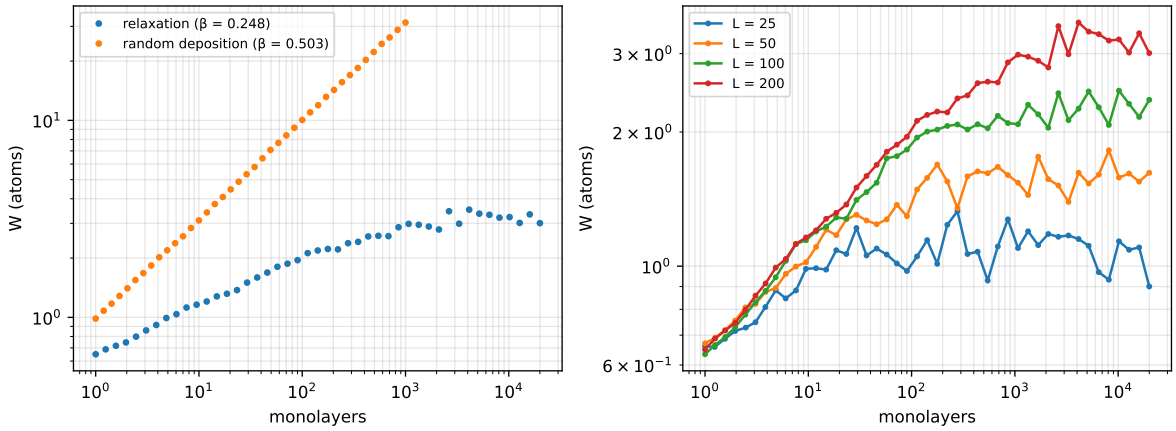


Figure 2: Left: relaxation roughens much more slowly than random deposition. Right: roughness saturates, and the plateau rises with substrate width L.

curves onto one (see `results/phase2_surface_relaxation/figure.png`). This is the Family–Vicsek scaling form (Family and Vicsek 1985), with exponents expected for Edwards–Wilkinson growth (Edwards and Wilkinson 1982).

Why Numba. In Phase 1 the order of arrivals did not matter, so atoms could be added in bulk with NumPy. Here each atom's destination depends on the surface it finds, so atoms must be processed one by one in a loop. Compiling that loop with Numba's `@njit` runs it at 3.72e+08 atoms/s instead of 2.91e+06, a **128× speed-up**. A test checks that both versions build exactly the same film from the same random numbers.

5 Phase 3: temperature and deposition rate

5.1 The physics

Between landings, surface atoms try to hop sideways. A hop succeeds with probability

$$P_{\text{hop}} = e^{-n\,E_{\text{bond}}/kT},$$

where $n \in \{0, 1, 2\}$ is the number of side neighbours at the atom's height. An isolated atom on a flat terrace hops freely; an atom tucked against a step is held more strongly. Real conditions map onto two simulation numbers:

- $\epsilon = E_{\text{bond}}/(kT)$: how sticky bonds are compared with the thermal energy.

- **Hops per landing:** $\nu_0 \exp[-E_{\text{diff}}/(kT)]/F$. the Arrhenius hop rate of a free atom divided by the arrival rate F. It counts how many moves an atom gets before the next layer buries it.

The barriers are illustrative (E_bond = 0.2 eV, E_diff = 0.8 eV, $\nu_0 = 10^{13}$ s^{-1}), so temperatures are meaningful relative to the model rather than for a specific metal.

5.2 Results

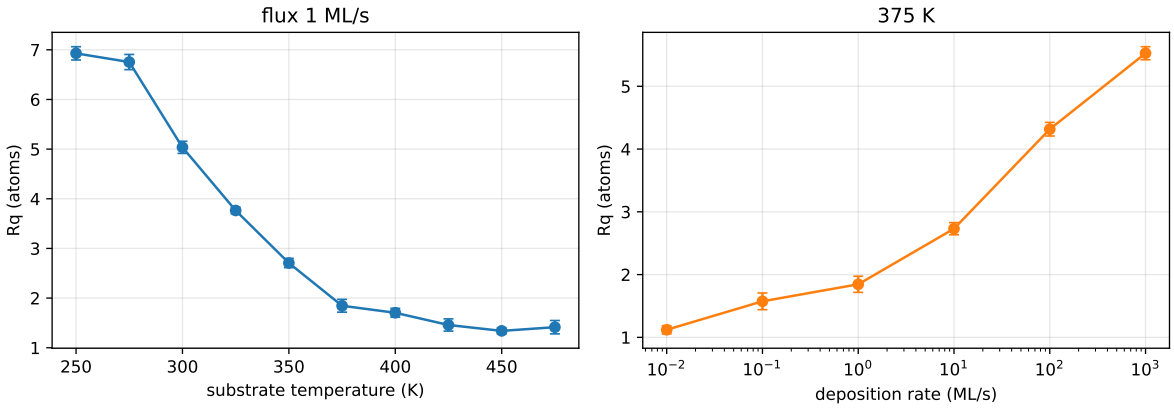


Figure 3: Roughness after 50 monolayers. Left: heating gives atoms more hops, so they find well-bonded sites. Right: at fixed temperature, faster deposition buries atoms before they can settle.

- Heating from 250 K to 450 K cuts roughness from 6.9 to 1.3 atoms.
- Slowing deposition 10^5-fold at 375 K cuts it from 5.5 to 1.1.
- **Temperature and rate are interchangeable** through the hop count. Every +40 K buys roughly 10× faster deposition at the same roughness. This is the trade-off fabs face between substrate heating (thermal budget) and throughput.
- **Bonds matter as much as mobility.** In the dimensionless model at 100 hops per landing, removing bonding entirely ($\epsilon = 0$) makes the film *much* rougher (W = 36). Atoms then wander uphill as easily as downhill. Even weak bonding ($\epsilon = 1$) gives W = 4.1: smoothing needs atoms that move *and* stay put once they find a good site.

5.3 Process window and recipe

A grid over temperature × rate (`results/phase3_process_window`) finds the smoothest film at **375 K and 0.01 ML/s**. That is the coolest temperature at which slow deposition still gives the maximum number of hops while bonds stay strong.

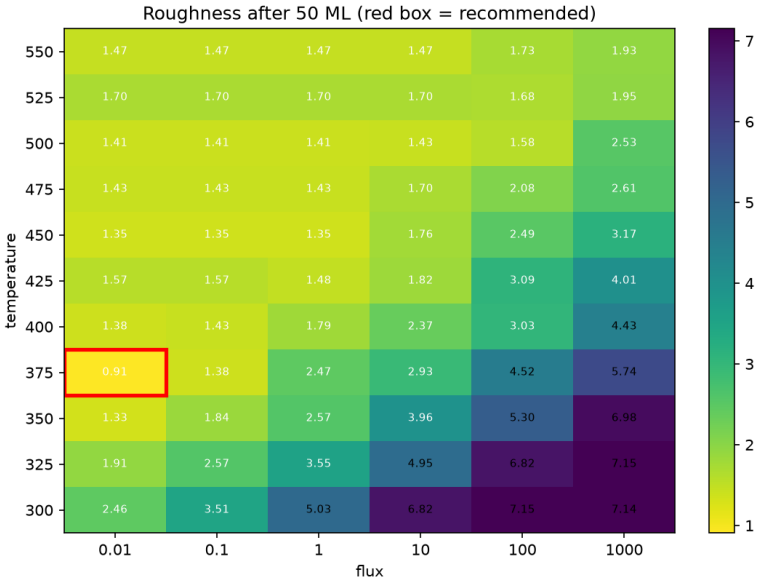


Figure 4: Roughness over the temperature × rate grid. The red box marks the POR.

6 Phase 4: filling a trench

6.1 The physics

This is a 2D side view of a wafer with a rectangular trench. Atoms fall straight down and **stick on first contact**, below or beside them. Atoms grazing the trench's top corners stick to the corner's side, so the film grows sideways from both corners as **overhangs**. If the overhangs meet before the bottom fills, they seal a void.

Temperature enters as in Phase 3: each new atom gets a number of hop attempts, succeeds with probability $e^{-n\epsilon}$ for n filled neighbours, and moves to the neighbouring empty cell with the most bonds. Hot atoms therefore drift into corners and along the trench floor. Voids are found by a **flood fill** from the open space above: any empty cell the flood cannot reach is sealed. (This model uses a weaker bond, E_bond = 0.1 eV. At 0.2 eV, bonded atoms in 2D barely move and every trench voids.)

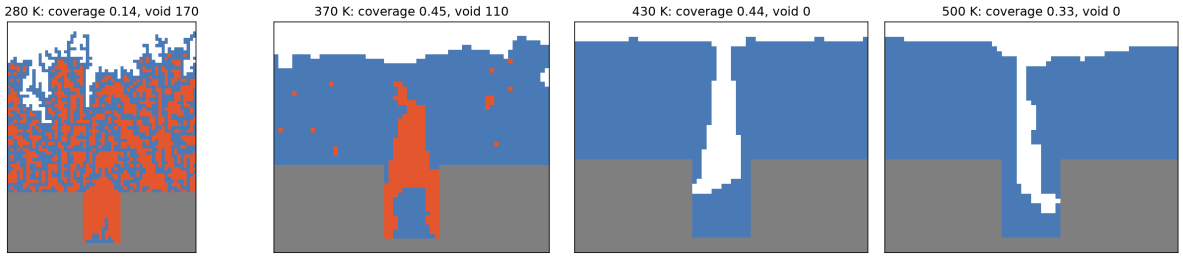


Figure 5: Cross-sections after 24 monolayers (red = void). Cold films are porous and seal immediately; at 370 K the film is dense but overhangs still trap a void; from about 430 K atoms reach the bottom and the trench fills, although a narrow seam can stay open to the surface (open space, so not counted as a void).

6.2 Temperature and width

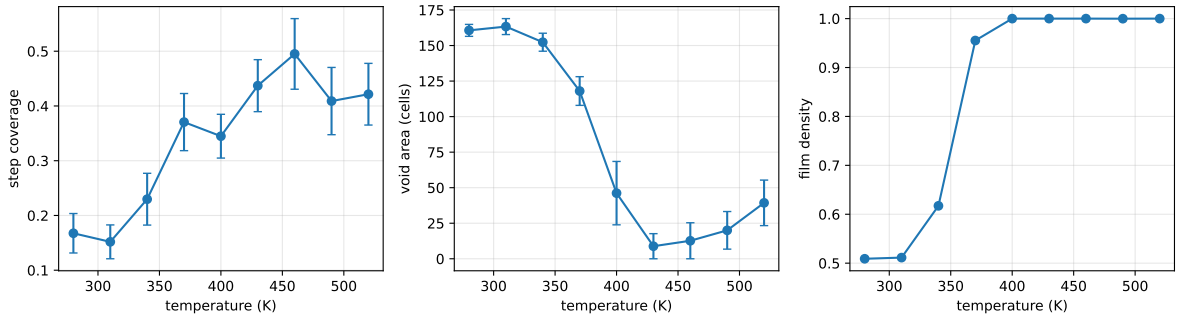


Figure 6: Trench 12 wide × 16 deep. Heating densifies the film and raises step coverage, and voids shrink sharply above about 400 K. Above about 460 K the void area creeps back up: hops are capped at 3000 per atom, so extra heat only weakens bonds.

The main lesson: **aspect ratio (depth ÷ width) decides first, temperature second.** In a deep, narrow trench the corners are close together, so the overhangs meet after only a few monolayers. That happens long before diffusing atoms can carry enough material to the bottom. Fabs face the same limit: line-of-sight deposition cannot fill high-aspect-ratio features, which is why they turn to reflow, atomic layer deposition or electroplating.

6.3 Process window and recipe

A grid over width × temperature × rate (`results/phase4_process_window`, 8 runs per cell) gives the POR: **16 × 16 trench, 420 K, 1 ML/s**. It is void-free on average with coverage 0.59. That is within noise of the best coverage found, at a temperature 100 K lower than the best cell.

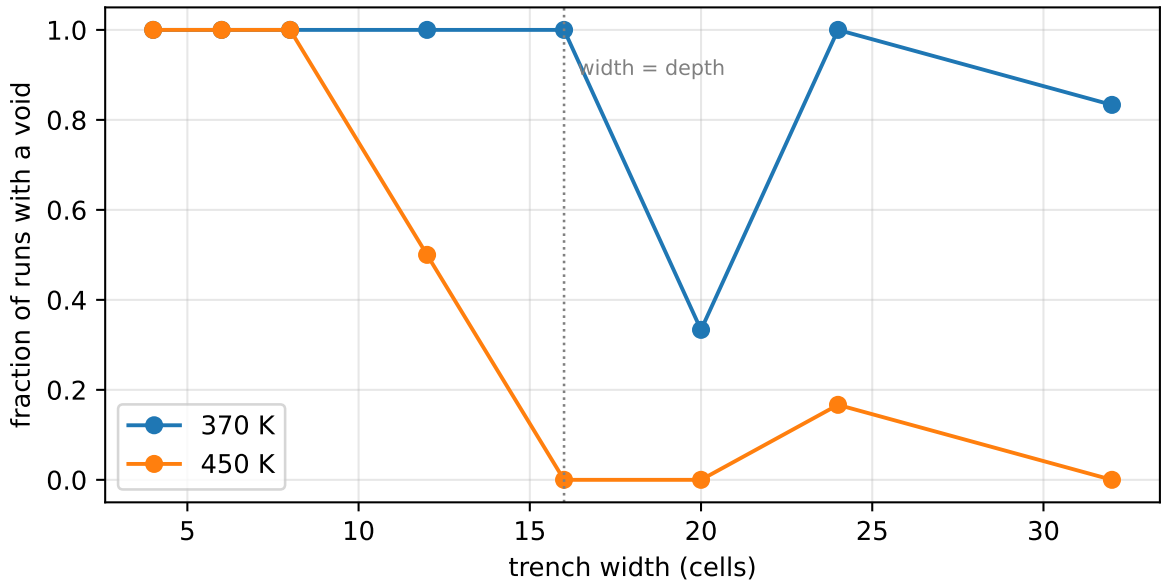


Figure 7: Fraction of runs with a trench void, by trench width (depth 16). Narrow, deep trenches void at any temperature; at 450 K trenches at least as wide as they are deep fill without a void in nearly every run (one run in six at width 24 trapped a 16-cell void).

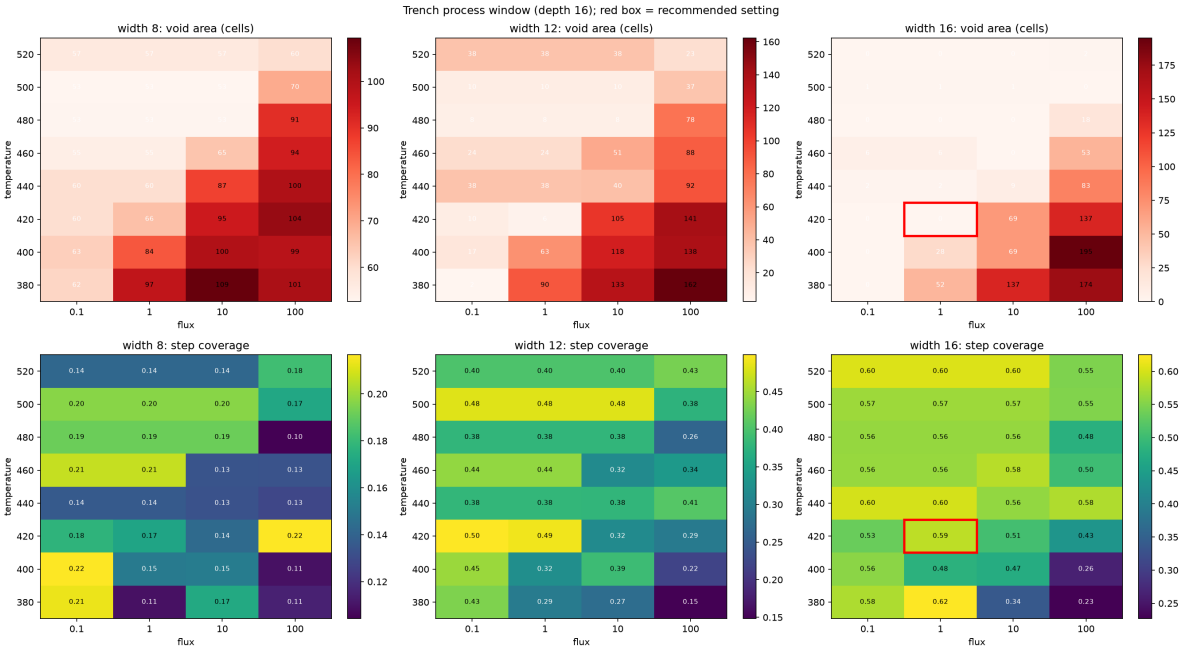


Figure 8: Void area (top) and step coverage (bottom) over the process window; the red box marks the POR.

7 From averages to wafers: yield

A spec applies to each wafer, not to an average. So the right question is *what fraction of single wafers pass?* Each recipe was run with 30 seeds exactly as the app runs it (`results/por_yield`). A flat film passes when $R_q \leq 2.0$ atoms at 100 ML. A trench passes when there are no sealed trench voids and step coverage is at least 0.50. The results below are empirical pass fractions over these seeds, not manufacturing yield estimates:

Table 1: Single-wafer pass rates over 30 seeds.

	module	recipe	passed	wafers	yield
0	thermal	POR	30	30	100%
1	thermal	375 K, 1 ML/s	2	30	7%
2	thermal	450 K, 1 ML/s	24	30	80%
3	trench	POR	24	30	80%
4	trench	460 K, 10 ML/s	16	30	53%
5	trench	370 K, 1 ML/s	9	30	30%

- The 1D POR passes every wafer. Its fast alternative (450 K, 1 ML/s) is 100× faster but passes only 80%.
- The trench POR passes about 80%. Failures are near misses: coverage just under 0.50, or a void of a few cells. Spread between wafers made with the same recipe is a real feature of growth, and it is why averages alone can mislead. A recipe that is void-free on average at 460 K and 10 ML/s passes only about half its wafers.

8 Discussion

The flat-film and trench experiments separate two mechanisms. Greater surface mobility gives atoms more opportunities to find well-bonded sites, reducing roughness in the thermal model. In a trench, however, the opening may close before enough material reaches the bottom. Improving mobility does not remove every geometric constraint, so smooth or dense films are not sufficient evidence of successful feature filling.

The distinction between averages and individual runs is also consequential. The trench reference has zero mean sealed-void area in its process-window sample, yet only 24 of 30 wafers pass the combined void and coverage criteria in the separate yield experiment. Selecting a recipe therefore requires both a process-window search and repeated evaluation against the full specification. The 30-seed pass fractions should be interpreted as finite simulation samples.

The numerical hop cap further limits extrapolation. Once diffusion attempts reach the cap, increasing temperature can weaken bonding without increasing the simulated mobility proportionally. The high-temperature upturn in void area should therefore be interpreted as a feature of the capped model, not as a general material law.

9 Limitations

- **Lattice units.** 1 cell = 1 atom. The nanometre values in the app assume 1 ML ≈ 0.3 nm and are illustrative only.
- **Illustrative energies.** E_bond and E_diff are not fitted to any material, so only relative temperatures are meaningful.
- **Capped hops.** Runs are capped at 10^4 hops per landing in 1D and 3000 per atom in 2D. The upturn in trench voids above about 460 K comes from this cap. A rejection-free kinetic Monte Carlo would remove it.
- **Simplified 2D motion.** Only the newest atom hops, and atoms arrive straight down. Real sputtering has an angular spread, which affects overhang growth directly.
- **Open seams.** A narrow seam that stays open to the surface is not counted as a void, by definition.

10 Conclusion

The simulation recovers the expected approximate roughening scalings of its validation models and connects thermal mobility to roughness and trench filling. Within the tested grids, trench aspect ratio is a stronger constraint than temperature alone. Repeated runs show that an average process window does not establish reliable single-wafer acceptance. The resulting reference recipes are useful for exploring these relationships in the simulator, while material-specific prediction would require calibrated energies, more realistic incident flux and diffusion dynamics without the present hop cap.

Appendix A: Interactive simulator

Try the live Streamlit application. Choose a process module, set temperature, deposition rate and geometry, then run a wafer and inspect its growth history. Compare the recipe against the process of record (POR), replay the deposition, or review lot history.



Figure 9: The simulator at inspection, using the supplied screenshot. A 12-cell-wide, 16-cell-deep trench at 375 K and 1 ML/s fails: 100 void cells and step coverage 0.42, against a requirement of zero void cells and coverage at least 0.50. The SEM-style cross-section is simulated, and its nanometre scale is illustrative.

This is an example of a failed recipe, distinct from the trench POR (16 × 16 cells, 420 K, 1

ML/s). It shows why high film density (0.99 here) alone does not establish successful trench filling.

Appendix B: Reproducibility

```
python3.12 -m venv .venv && source .venv/bin/activate && pip install -r requirements.txt
for p in 1 2 3 4; do PYTHONPATH=src python scripts/run_phase$p.py; done
PYTHONPATH=src python scripts/run_por_yield.py
cd report
QUARTO_PYTHON=../.venv/bin/python quarto render report.qmd --to all
QUARTO_PYTHON=../.venv/bin/python quarto render showcase.qmd --to all
```

Every result folder holds `settings.json` with the inputs, seed, fitted values and library versions.

References

Edwards, S. F., and D. R. Wilkinson. 1982. "The Surface Statistics of a Granular Aggregate." *Proceedings of the Royal Society of London. A* 381 (1780): 17–31. https://doi.org/10.1098/rspa.1982.0056.

Family, Fereydoon, and Tamás Vicsek. 1985. "Scaling of the Active Zone in the Eden Process on Percolation Networks and the Ballistic Deposition Model." *Journal of Physics A: Mathematical and General* 18 (2): L75–81. https://doi.org/10.1088/0305-4470/18/2/005.